Fair for Whom?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected performance-comparison audit.

Explain the shape-test fraction, compare conditions, inspect missing outcomes and distinguish aggregation from equal-group weighting.

Keep one habit: Keep correct counts, assessed counts, missing outcomes and conditions visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Recommendations. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Prediction** Is Not Fact. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Missing and Unbalanced Examples

Recommendations

Prediction Is Not Fact

W1 · Explain the performance fraction

> Use the supplied shape-test counts and explain every part of the denominator.

Your evidence record

- Shared shape-test reference for W1–W2: fixed versions A and B label the same 12 previously unseen cards with checked reference labels: 8 bright and 4 dim. No outcomes are missing. A is correct on 8 bright and 1 dim; B on 6 bright and 2 dim. Keep this case separate from the 16-card badge test and the independent sound test.

- For version A in the lesson, write correct and assessed counts, simplify the **accuracy** fraction and express it as a percentage. Explain what its **denominator** includes and why 9/9 would be misleading.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compare the total and each condition

> Use the same cards and scoring rule for both versions.

Your evidence record

- Compare A and B on the shared 12-card test and on its bright and dim groups. State the leader in each and explain why the overall result does not imply that one version wins in every condition.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Report unassessed outcomes separately

> Distinguish assessed accuracy from an all-selected-case claim.

Your evidence record

- C has 9 correct and 1 incorrect assessed outcome, with 2 other selected cases unassessed. Write its assessed **accuracy** and missing count. A reviewer reports 9/12 as assessed **accuracy**; another reports 90% on all 12.

  Explain the problem with each label.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Audit an unequal-group average

> Add counts before dividing; label any alternative group metric explicitly.

Your evidence record

- A separate test has 2 correct out of 2 assessed in group P and 4 correct out of 8 assessed in group Q. Calculate combined **accuracy** as a fraction and percentage. A chart reports 75% by averaging the group percentages.

  Explain why that is not the **accuracy** over all assessed cases.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Compare a new shared sound test

> Keep one first attempt before hints or feedback.

Read and explain independently

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Compare a new shared sound test

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Test the sorter across relevant item conditions and users; never infer sensitive identity to create a **fairness** check.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Compare totals and conditions for Fractions and performance comparisons. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Fractions and performance comparisons. Proposed explanation: “Only correct cases belong in the **denominator**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A voice system succeeds on 8/10 quiet and 4/10 noisy recordings.

Compare and propose the next investigation.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A helper succeeds on 4 of 4 clear voices and 1 of 4 quiet voices. Who is less well served? What **evidence** should a repair preserve?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 23 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

System A succeeds on 6 of 8 quiet tests and 3 of 4 noisy tests. Is its noisy success rate lower? Show fractions and percentages. What can these small tests not establish?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Versions X and Y label the same 20 previously unseen clips: 12 quiet and 8 noisy, with no missing outcomes. X is correct on 10 quiet and 2 noisy clips; Y on 9 quiet and 5 noisy clips.

Calculate each combined fraction and percentage, identify the overall leader and compare the leaders within each group. Explain why an overall result cannot guarantee the next clip's outcome.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

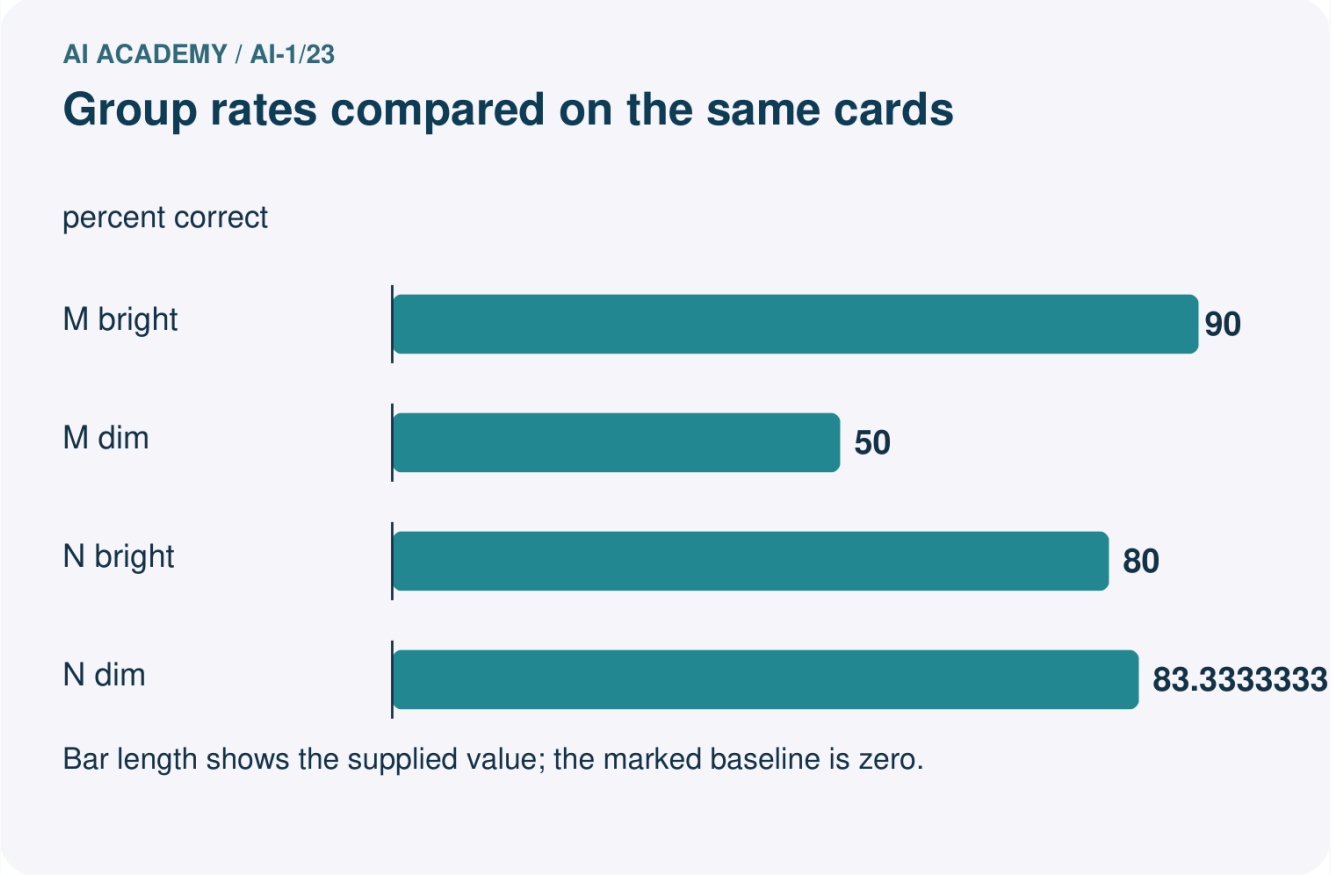
> Keep the explanation and the support you used.

Use this when helpful

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/23

Fair for Whom?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Bright group

M:9/10=90%; N:8/10=80%.



2. Dim group

M:3/6=50%; N:5/6≈83.3%.



3. Combined counts

M:(9+3)/16=75%; N:(8+5)/16=81.25%.



4. Discuss trade-off

N improves total and dim results but loses one bright success.

CHANGE ONE THING

Give bright and dim groups equal weight as an explicitly different metric.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Which group loses performance under N, and can total improvement erase that fact?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional badge reader is evaluated on bright and dim cards with known reference labels.
- Which labelled step matches this detail? N improves total and dim results but loses one bright success.
- 1. Dim group
- 2. Combined counts
- 3. Bright group
- 4. Discuss trade-off
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Discuss trade-off N improves total and dim results but loses one bright success. Adding correct counts and **denominator**s weights each tested card once.

Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**.

Changed-case question

- Changed-case practice: Give bright and dim groups equal weight as an explicitly different metric. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. M scores **70%**; N scores about 81.7%.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: M scores **70%**; N scores about 81.7%. This averages group rates and is not the original 16-card agreement rate.

Repair and save your work

Review Missing and Unbalanced Examples (ai-1/22). Goal: Name the target collection, selection list and cases actually assessed. Short practice: Would testing only dim plain cards close every current gap?

Point to the target groups.. Return to this week and explain what you changed.

Review Recommendations (ai-1/20). Goal: Apply eligibility requirements before ranking candidates by preferences. Short practice: Why does the recommendation change without changing anyone’s preferences?.

Return to this week and explain what you changed.

Review **Prediction** Is Not Fact (ai-1/17). Goal: Separate a recorded **prediction** from the **observation** used to check it. Short practice: Which field changes the **observation** label, and which field changes the model **prediction**?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Fair for Whom?: the case

AI ACADEMY / AI-1/23 · 01 · LOOK AT THE FACTS

Fair for Whom?: the case

The situation

A fictional badge reader is evaluated on bright and dim cards with known reference labels.

Supplied evidence

Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.
Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

By the end, I can compare group fractions and combine unequal-sized groups correctly.

Follow the labelled picture with the matching explanation. Open the vector diagram

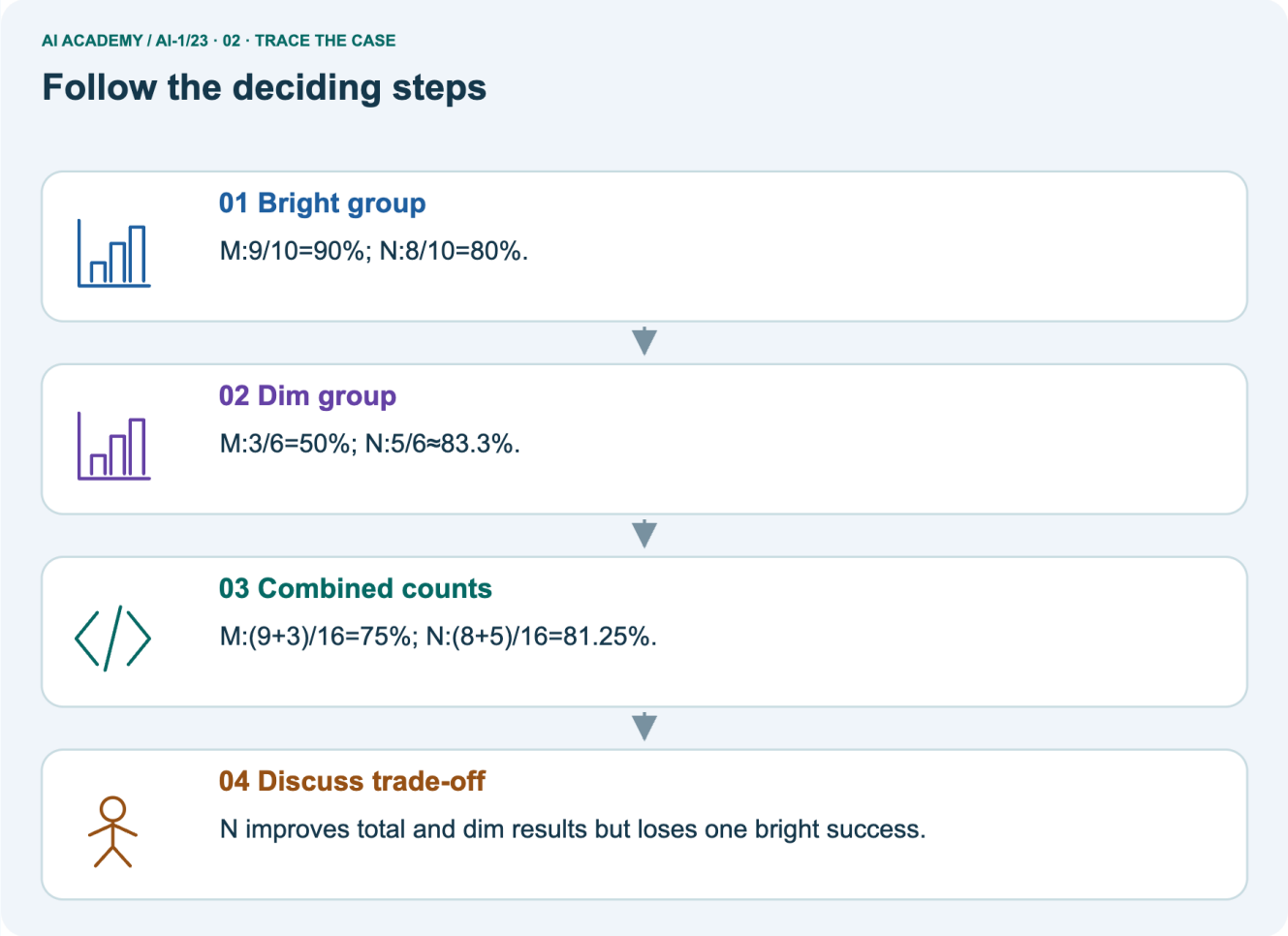
A fictional badge reader is evaluated on bright and dim cards with known reference labels.
Supplied facts: Version M is correct on 9 of 10 bright cards and 3 of 6 dim cards.

Version N uses the same cards: 8 of 10 bright and 5 of 6 dim correct.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



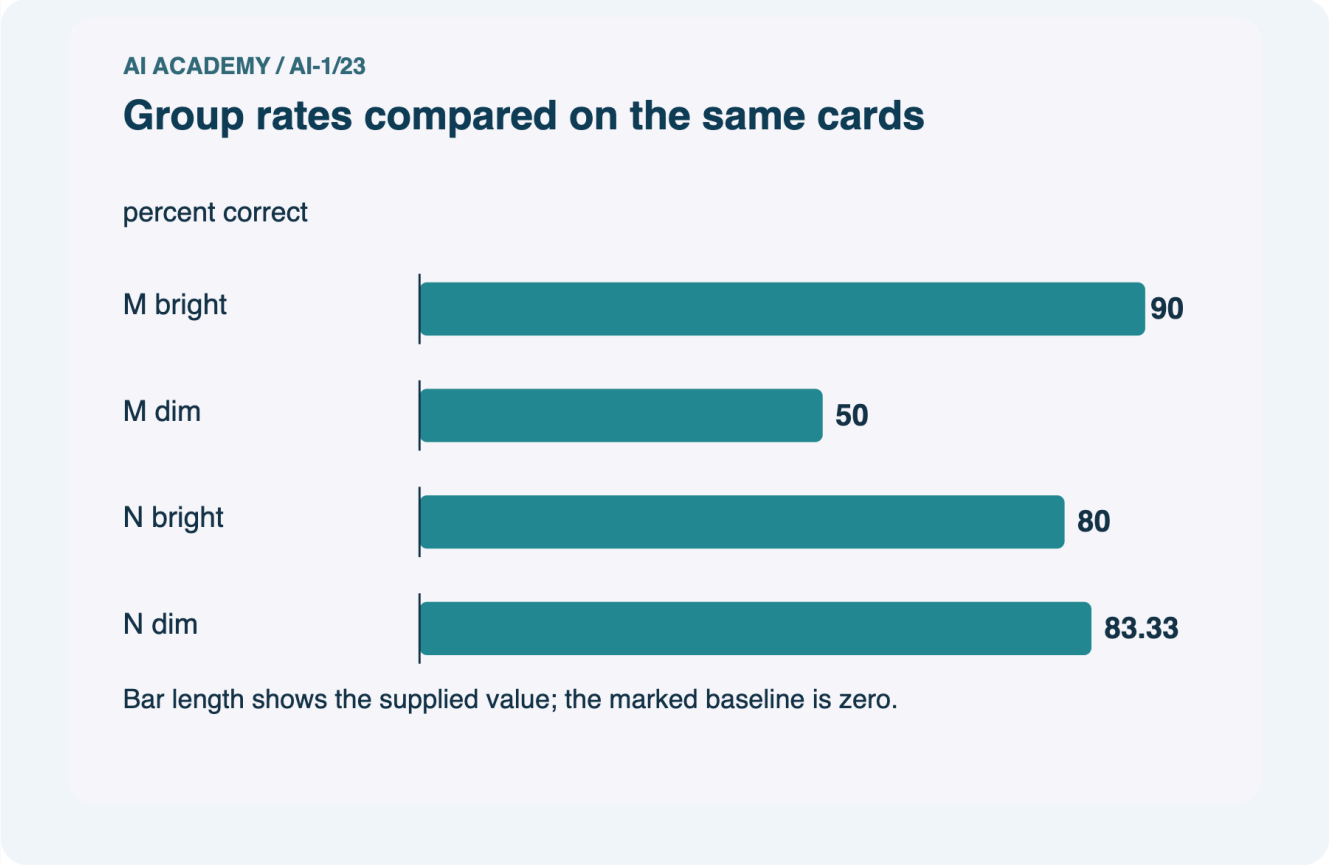
Follow the labelled picture with the matching explanation. Open the vector diagram

Bright group: M:9/10=90%; N:8/10=**80%**. → Dim group: M:3/6=50%; N:5/6≈83.3%. → Combined counts: M:(9+3)/16=75%; N:(8+5)/16=81.25%. → Discuss trade-off: N improves total and dim results but loses one bright success.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Group rates compared on the same cards



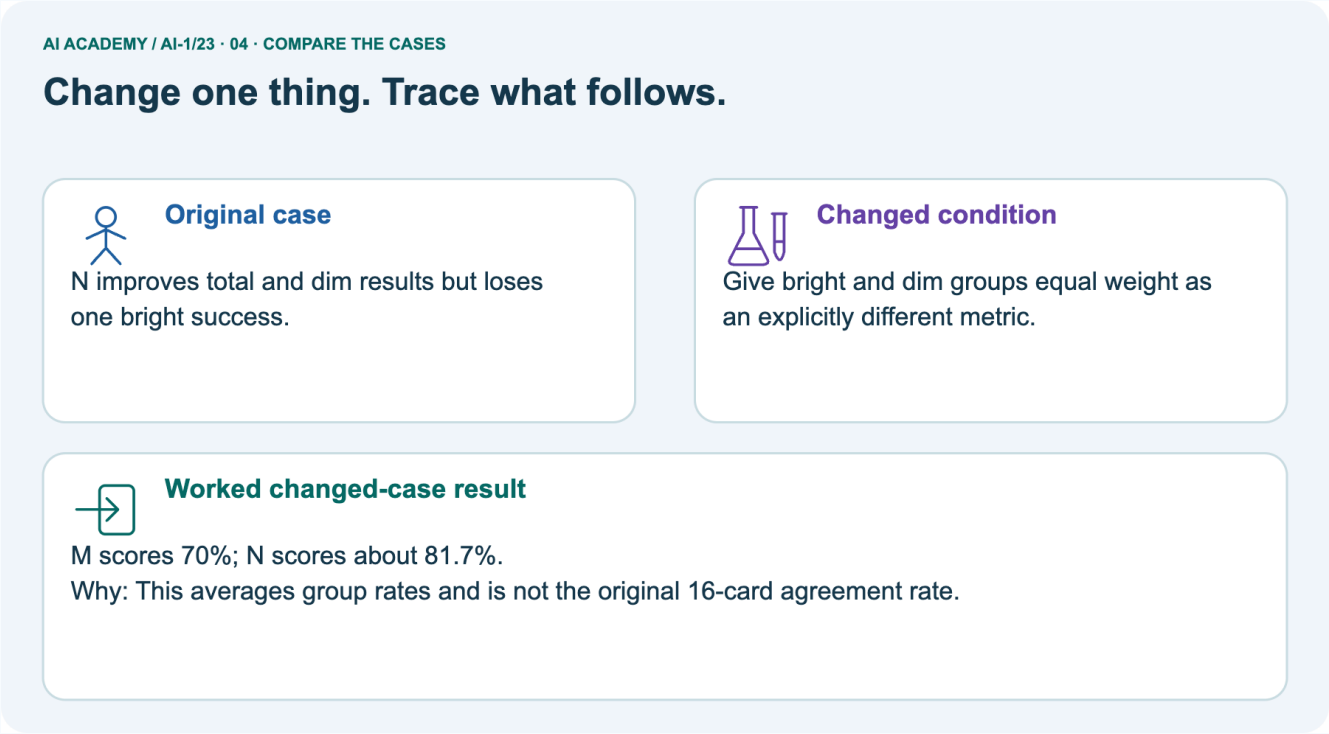
Follow the labelled picture with the matching explanation. Open the vector diagram

Group rates compared on the same cards. Units: percent correct. M bright: 90. M dim: 50. N bright: 80. N dim: 83.33333333333333.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

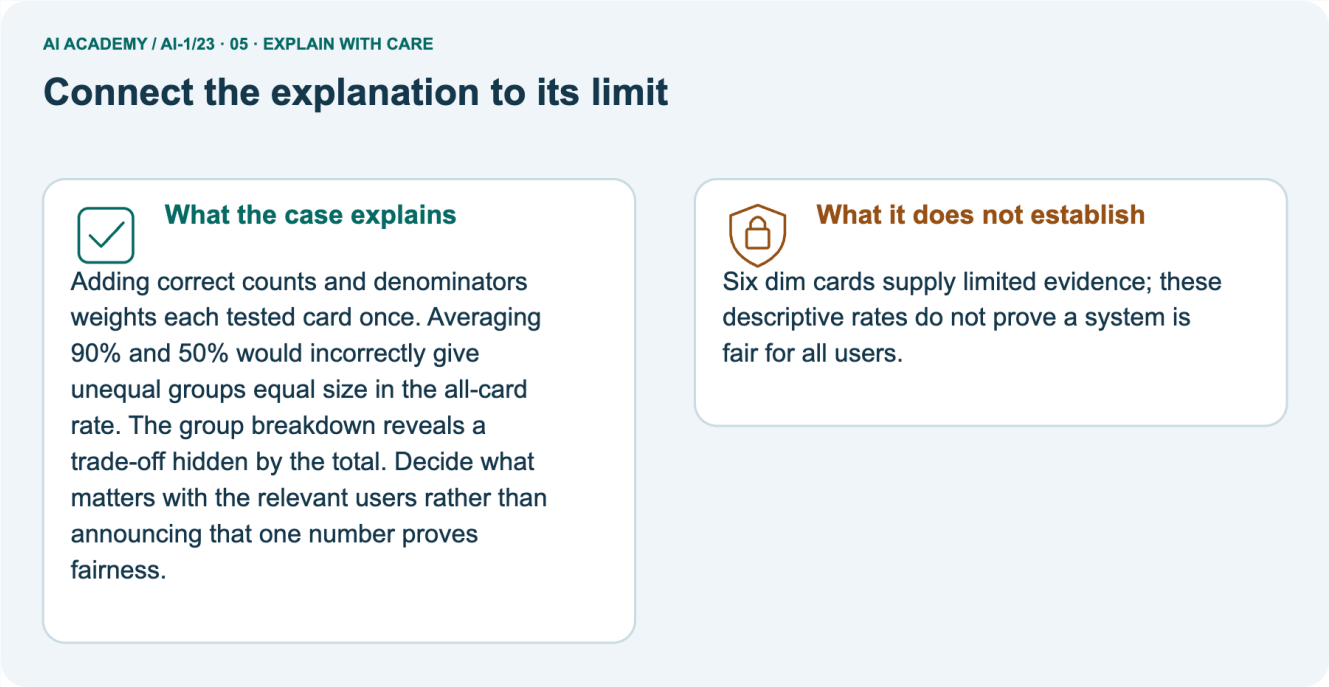
Original case: N improves total and dim results but loses one bright success. Changed condition: Give bright and dim groups equal weight as an explicitly different metric. Expected result: M scores **70%**; N scores about 81.7%.

Reason: This averages group rates and is not the original 16-card agreement rate.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

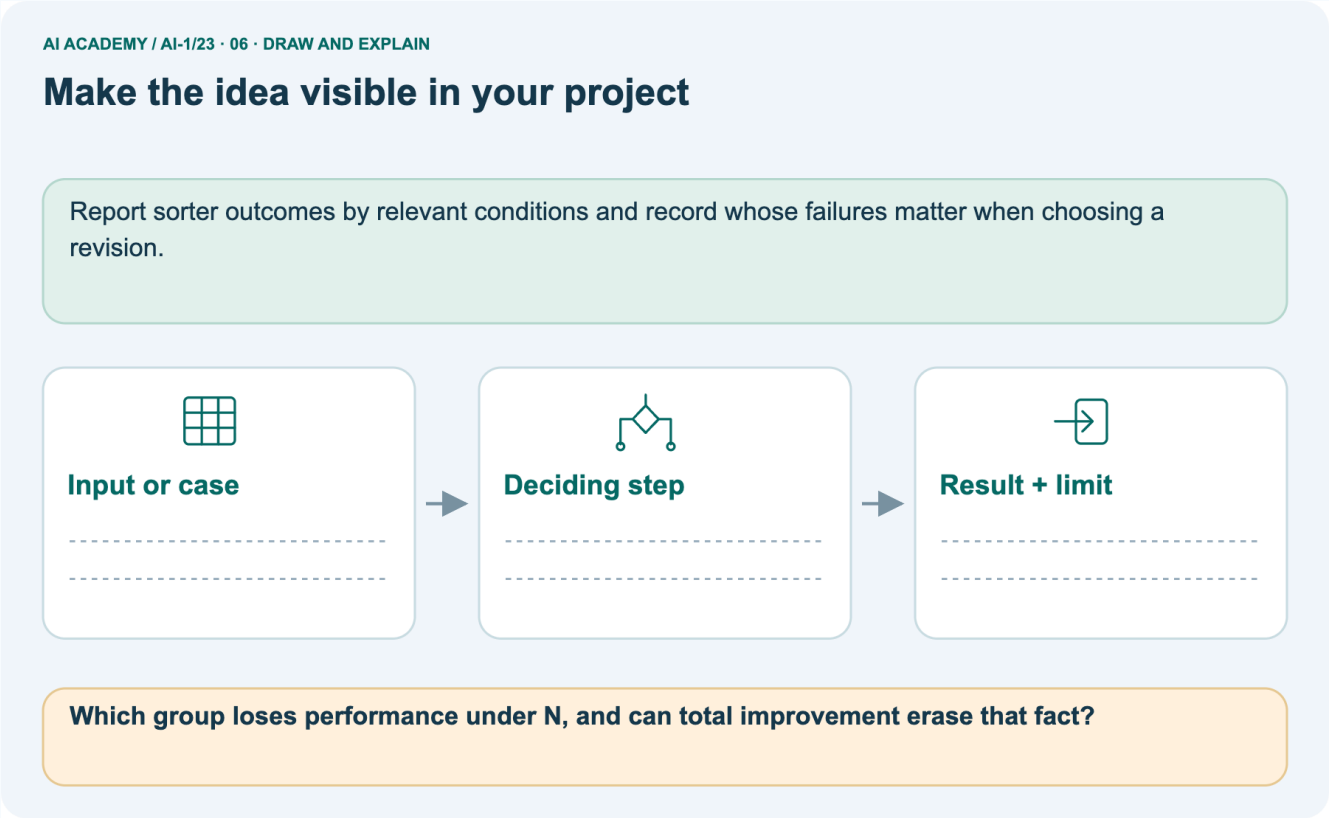
Adding correct counts and **denominator**s weights each tested card once. Averaging 90% and 50% would incorrectly give unequal groups equal size in the all-card rate. The group breakdown reveals a trade-off hidden by the total.

Decide what matters with the relevant users rather than announcing that one number proves **fairness**. Limit: Six dim cards supply limited **evidence**; these descriptive rates do not prove a system is fair for all users.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Report sorter outcomes by relevant conditions and record whose failures matter when choosing a revision. Draw your **input**, deciding step, result and limit.

Supported question: Which group loses performance under N, and can total improvement erase that fact?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Fair for Whom?: the case

02. Follow the deciding steps

03. Group rates compared on the same cards

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.